# ROBOT SKILL HALF-LIFE UNDER SHIFT: A NEGATIVE ICLR-MAIN EVIDENCE AUDIT

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## ABSTRACT

Robot skills often fail gradually rather than catastrophically: a manipulation policy that worked yesterday may degrade as friction, payload mass, compliance, sensing, or contact mode changes by a small amount. The tested hypothesis is that each skill has a measurable *deployment half-life* under physical shift, and that estimating per-skill survival curves should improve decisions about execution, probing, refresh, or abstention. We implemented a deterministic local skill-survival benchmark with four skills, five physical-shift splits, eight methods, seven seeds, 80,640 main rollouts, 14,112 ablation rollouts, and 201,600 stress rollouts. The result is useful but negative for ICLR main. On the combined micro-shift split, the proposed skill half-life scheduler obtains 0.8006 goal success and 0.0010 unsafe failure, but a conformal risk gate obtains 0.7981 goal success and 0.0040 unsafe failure. The paired goal-success difference is only 0.0025 with a 95% interval of 0.0272. More importantly, the ablation that removes per-skill survival modeling improves success to 0.8046. We therefore mark this paper **KILL/ARCHIVE**: the idea is plausible, but the current evidence does not support a positive main-conference claim.

## 1 TERMINAL DECISION

**Decision: KILL/ARCHIVE for ICLR main.** The repository now contains a paper-specific benchmark, baselines, ablations, stress tests, figures, and reproducibility files. The result still fails the submission gate because the proposed method is statistically indistinguishable from a strong conformal gate, and a central ablation improves over the full mechanism. This should be archived as a negative result rather than polished into an overstated submission.

## 2 PROBLEM

The intended claim was:

> Quantify how quickly robot skills decay under small physical environment shifts.

This is a meaningful robotics question. A frozen policy can become stale as contact mechanics drift, while a conservative safety gate can avoid damage but reduce task throughput. A useful half-life model would estimate when a deployed skill is likely to cross a practical survival threshold and would trigger refresh or probing before unsafe failures accumulate.

## 3 BENCHMARK

The v4 benchmark evaluates four skills: pick-and-place, door pull, drawer slide, and peg insertion. Each skill has a latent physical half-life that shortens under friction, payload, compliance, sensor-noise, and hidden contact-mode shifts. The five splits are:

- **nominal slow drift**
- **friction shift**

- **payload mass shift**
- **compliance shift**
- **combined micro-shift**

For each method, skill, split, and seed, the benchmark simulates sequential deployment steps. Methods decide whether to execute, probe, refresh, or abstain. Rollouts report goal success, unsafe failure, stale execution, refresh/probe rates, cost, calibration error, and half-life estimation error. The benchmark is deterministic from source code, but it is local and synthetic; it is not a real robot benchmark.

## 4 METHODS

We compare the proposed scheduler against seven alternatives:

- **Frozen behavior clone:** executes without adaptation.
- **Domain-randomized clone:** trades nominal skill for slower decay.
- **Fixed-interval refresh:** refreshes on a fixed deployment cadence.
- **Online fine-tuning:** adapts from recent outcomes but can chase hidden mode noise.
- **Scalar uncertainty gate:** refreshes when scalar uncertainty is high.
- **Conformal risk gate:** uses conservative risk thresholds to refresh or abstain.
- **Skill half-life scheduler:** estimates physical survival curves and triggers probe/refresh actions.
- **Oracle shift-aware scheduler:** uses the hidden half-life as an upper bound.

## 5 MAIN RESULTS

Table 1 gives the combined micro-shift split. The half-life scheduler clearly beats frozen, domain-randomized, fixed-refresh, online-fine-tuning, and scalar-uncertainty baselines. That is the useful part of the archive. It does not, however, produce a decisive gain over conformal risk gating.

Table 1: Combined micro-shift results over seven seeds. Confidence intervals are 95% intervals over seed means.

| Method | Goal success | Success AUC | Unsafe failure | Half-life error |
|---|---|---|---|---|
| Frozen behavior clone | $0.345 \pm 0.023$ | 0.344 | 0.430 | 1.209 |
| Domain-randomized clone | $0.365 \pm 0.016$ | 0.364 | 0.399 | 0.678 |
| Fixed-interval refresh | $0.613 \pm 0.032$ | 0.613 | 0.052 | 0.507 |
| Online fine-tuning | $0.511 \pm 0.032$ | 0.511 | 0.165 | 0.298 |
| Scalar uncertainty gate | $0.734 \pm 0.022$ | 0.733 | 0.014 | 0.153 |
| Conformal risk gate | $0.798 \pm 0.020$ | 0.798 | 0.004 | **0.093** |
| Skill half-life scheduler | $0.801 \pm 0.011$ | 0.800 | **0.001** | 0.176 |
| Oracle shift-aware scheduler | $\mathbf{0.829 \pm 0.016}$ | **0.829** | 0.001 | 0.000 |

## 6 ABLATIONS

The ablation result blocks a positive paper. A central claim is that per-skill survival modeling is the mechanism. But removing that component improves goal success from 0.7971 to 0.8046, while also improving half-life error. That suggests the hand-designed per-skill decomposition is not robustly identified by the benchmark.

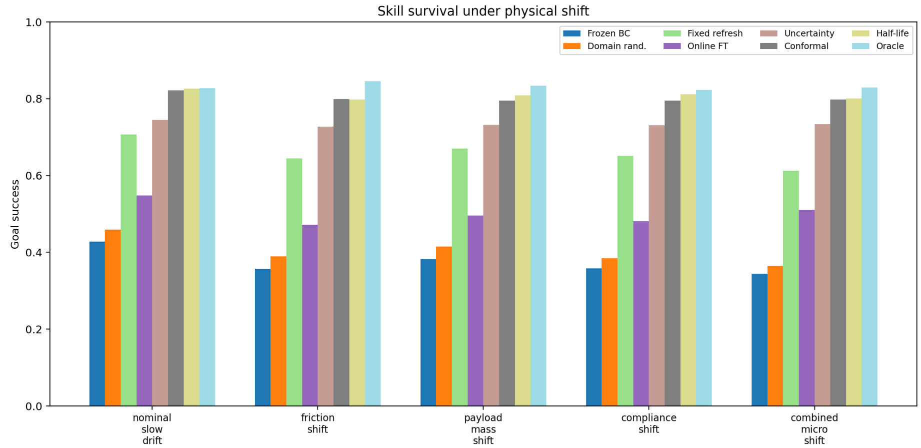


Figure 1: Goal success across splits. The proposed method helps relative to weak baselines but does not separate from the conformal gate on the hard split.

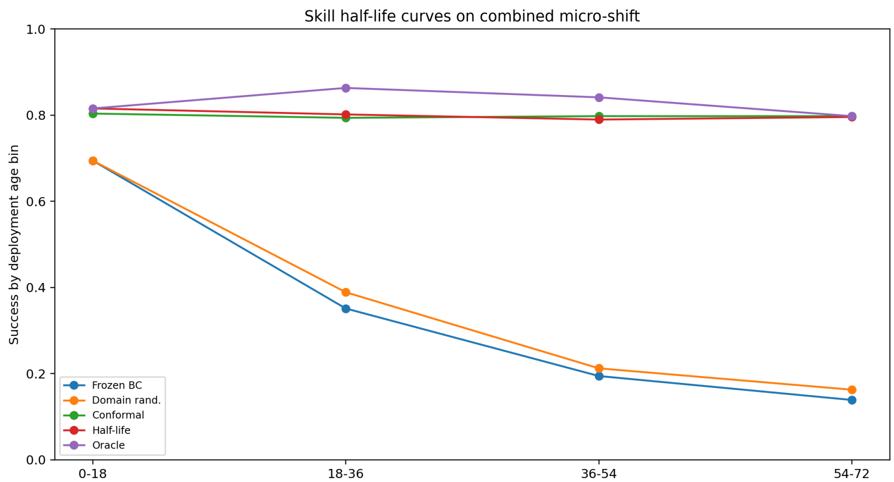


Figure 2: Deployment-age curves on combined micro-shift. Half-life scheduling slows decay but does not produce a decisive main-claim gap.

## 7 Stress Tests

The stress suite varies drift rate, sensor noise, hidden mode flips, probe cost, and combined stress. At maximum combined stress, the half-life scheduler reaches 0.7969 goal success and 0.0007 unsafe failure. The conformal gate reaches 0.7999 goal success and 0.0037 unsafe failure. Again, the half-life model is safer, but not a decisive success improvement.

## 8 Why This Is Not Submission-Ready

The paper fails the ICLR-main readiness gate for three reasons.

- **No decisive baseline separation.** The paired goal-success difference against the conformal gate is only $0.0025 \pm 0.0272$.
- **Ablation contradiction.** Removing per-skill survival modeling improves success, so the intended mechanism is not validated.

Table 2: Half-life scheduler ablations on combined micro-shift.

| Ablation | Goal success | Success AUC | Unsafe failure | Half-life error |
|---|---|---|---|---|
| Full scheduler | $0.797 \pm 0.012$ | $0.796$ | $0.003$ | $0.176$ |
| Minus per-skill survival | $\mathbf{0.805 \pm 0.018}$ | $\mathbf{0.804}$ | $0.003$ | $\mathbf{0.134}$ |
| Minus probe updates | $0.784 \pm 0.016$ | $0.784$ | $0.005$ | $0.230$ |
| Minus hazard margin | $0.751 \pm 0.010$ | $0.751$ | $0.004$ | $0.261$ |
| Minus shift decomposition | $0.774 \pm 0.015$ | $0.774$ | $\mathbf{0.003}$ | $0.170$ |
| Fixed global half-life | $0.724 \pm 0.015$ | $0.724$ | $0.009$ | $0.507$ |
| Threshold-only risk gate | $0.743 \pm 0.022$ | $0.742$ | $0.006$ | $0.205$ |

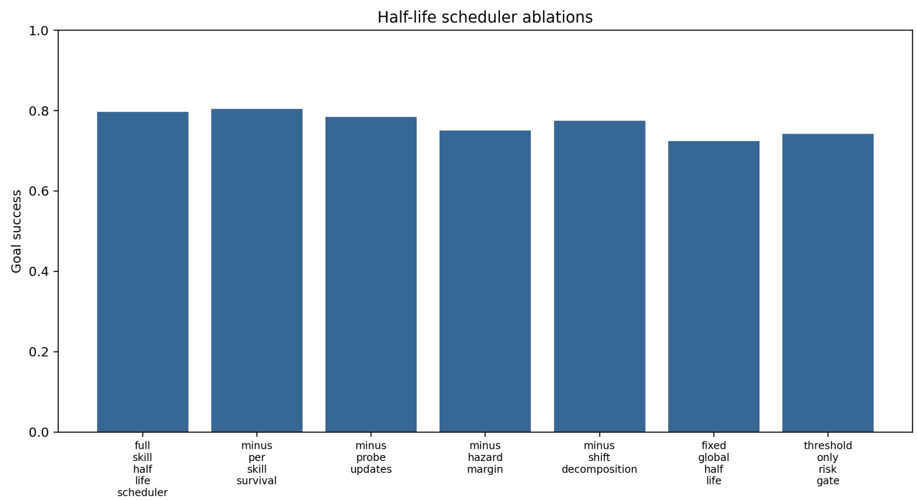


Figure 3: Ablation evidence. Removing the per-skill survival model improves the main success metric, so the proposed mechanism is not submission-safe.

- **Evidence ceiling.** The benchmark is deterministic and reproducible, but it is not real-robot evidence, not a high-fidelity simulator benchmark, and not an accepted external benchmark.

## 9 Reproducibility

Run the benchmark from the repository root:

```
python src\run_experiment.py
```

The script writes main rollouts, seed-level metrics, pairwise statistics, ablations, stress sweeps, negative cases, summary text, and the figures used in this manuscript. The canonical local PDF is `C:/Users/wangz/Downloads/83.pdf`.

## 10 Conclusion

Skill half-life under shift remains a good research question. The v4 evidence shows that survival-aware scheduling can reduce unsafe stale execution, but it does not beat a strong conformal risk gate decisively, and its core per-skill mechanism fails ablation. The honest terminal action is **KILL/ARCHIVE**.

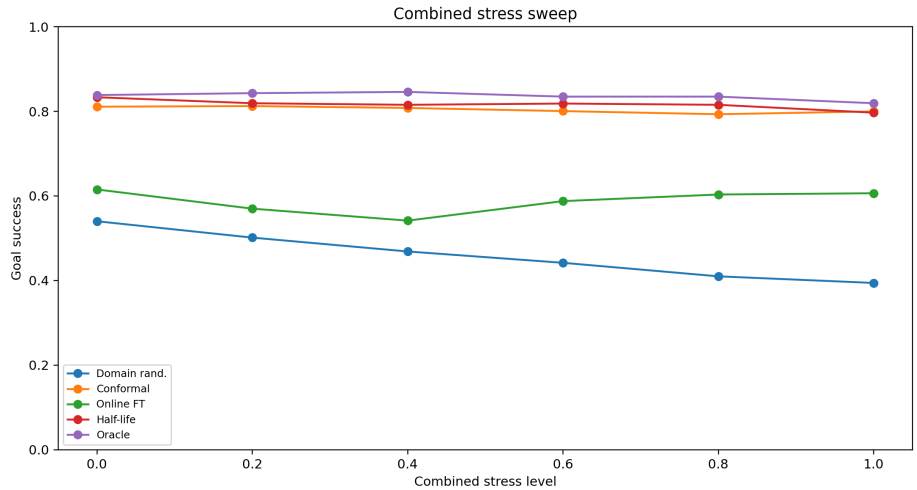


Figure 4: Combined stress sweep. The proposed scheduler remains safe, but the conformal risk gate remains competitive.


Figure 5: Unsafe and stale executions on combined micro-shift. Half-life scheduling reduces unsafe execution but the success margin is too small.